Biology Mock

Answers and Marking Scheme

B: independent mark (1 mark each) M: method A: answer (with unit) C: communication

Underlined words are key words. "/" shows an acceptable alternative; for "any n", only the first n points are marked; contradictory answers score no marks. For revision only; other valid answers should also be credited.

Section A Answers

1	2	3	4	5	6	7	8	9	10	11	12
C	A	B	B	A	B	D	C	C	D	A	D
13	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
B	D	A	B	C	C	A	B	D	C	D	C
25	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	**36**
A	B	A	B	C	B	B	C	A	A	A	A

1. C

2. A

3. B

4. B

5. A

6. B

7. D

8. C

9. C

10. D

11. A

12. D

13. B

14. D

15. A

16. B

17. C

18. C

19. A

20. B

21. D

22. C

23. D

24. C

25. A

26. B

27. A

28. B

29. C

30. B

31. B

32. C

33. A

34. A

35. A

36. A

Section B Marking Scheme

1. (3 marks)

(a) (3)

B1 P: nervous; the response starts very quickly (0.04 s) / lasts only 0.3 s (1)

B1 Q: hormonal; the response starts only after 5 days / lasts for months (1)

B1 R: hormonal; the response starts only after 20 min / lasts 2 h (1)

〔max. 3〕

Do not accept：只寫「快」「慢」而不引用表中數字；只寫協調方式而沒有數據

Common error：只寫協調方式，沒有引用表中具體數字；每行須方式和數據都正確才得分。

2. (3 marks)

(a) (1)

B1 substitution (1)

〔max. 1〕

Do not accept：誘發突變（未按鹼基序列的改變作答）；點突變（未寫置換）；缺失／插入

Common error：寫成「誘發突變」，只答了成因，沒有按序列的改變作答。

(b) (1)

B1 the mutations occur only in body (somatic) cells of the nasopharynx, not in the cells that produce gametes (in the testes / ovaries), so they cannot be passed to the children through gametes (1)

〔max. 1〕

Do not accept：突變是隱性的；只寫「不是遺傳病」

Common error：只寫「後天的突變不會遺傳」，沒有指出體細胞與配子的分別。

(c) (1)

B1 glycine is replaced by serine, changing the shape / function of the protein, so cells cannot stop dividing and divide uncontrollably to form a tumour (1)

〔max. 1〕

Common error：只寫「蛋白質改變」，沒有連繫到細胞週期失控。

3. (7 marks)

(a) (1)

B1 (infolded) inner membrane / cristae (1)

〔max. 1〕

Also accept：嵴；線粒體內膜的摺疊

Do not accept：外膜；基質

(b) (2)

B1 P has more folds of the inner membrane, providing a larger surface area for more enzymes of oxidative phosphorylation (1)

B1 heart muscle cells contract continuously and need much ATP, while fat cells mainly store fat and need less ATP (1)

〔max. 2〕

Common error：只描述 *P* 有較多摺疊，沒有比較兩種細胞對 *ATP* 的需求。

(c)(i) (1)

B1 glycolysis (forming pyruvate) (1)

〔max. 1〕

Also accept：葡萄糖分解為丙酮酸鹽的階段

Do not accept：克雷伯氏循環；缺氧呼吸（未寫階段）

(c)(ii) (3)

B1 pyruvate cannot enter the mitochondria and accumulates in the cytoplasm (1)

B1 pyruvate is converted to lactic acid in the cytoplasm, regenerating NAD (1)

B1 so that glycolysis can continue to produce ATP (1)

B1 lactic acid diffuses from the cells into the blood (1)

〔max. 3〕

Common error：只寫「進行缺氧呼吸」，沒有解釋丙酮酸鹽的去向及 *NAD* 的再生。

4. (8 marks)

(a) (2)

B1 exercising skeletal muscle; the lactic acid concentration in blood leaving Y rises sharply (3.6→7.8 mmol dm^{-3}), showing anaerobic respiration in Y (1)

B1 oxygen falls sharply from 100 to 28 / carbon dioxide rises from 100 to 131, showing a very high rate of respiration in Y (1)

〔max. 2〕

Also accept：肌肉／腿部肌肉（須配合乳酸或氧的證據）

Do not accept：沒有寫出器官名稱；以葡萄糖輕微下降作唯一證據

Common error：器官名稱錯誤則兩分都不給；只寫「氧下降」而沒有數據。

(b) (3)

B1 at the arterial end, blood pressure (4.3 kPa) is greater than the effect of plasma proteins drawing water back (3.3 kPa) (1)

B1 water and small solutes (e.g. glucose, amino acids) are forced out through the capillary wall to form tissue fluid (1)

B1 plasma proteins are too large to pass out and remain in the capillary (1)

B1 at the venous end, blood pressure (1.6 kPa) is less than 3.3 kPa, so water returns into the capillary by osmosis (1)

〔max. 3〕

Common error：沒有引用圖中數值比較兩種壓力。

(c) (3)

B1 more tissue fluid is formed at the arterial end than returns to the capillary at the venous end (1)

B1 the excess tissue fluid is normally drained away by lymph vessels as lymph (1)

B1 when lymph vessels are blocked, tissue fluid accumulates in the tissues, causing swelling (1)

〔max. 3〕

Common error：只寫「淋巴不能流動」，沒有解釋組織液的來源及積聚。

5. (9 marks)

(a) (3)

B1 cells lacking U show almost no haemolysis in the first 50 s, while normal cells are all haemolysed within 40 s (1)

B1 normal cells reach 50% haemolysis at about 15 s and 100% at about 40 s; cells lacking U reach 50% only at about 125 s and nearly 100% at about 300 s (1)

B1 the rate of haemolysis of normal cells is much faster (about 8 times) (1)

〔any 3；max. 3〕

Also accept：缺少 U 的一組要約 300 s 才接近全部溶血，正常組約 40 s；正常組達 50% 溶血所需時間約為缺少 U 一組的 1/8

Do not accept：只寫「正常細胞較快」而沒有數據；兩組最終都全部溶血（這是相同之處，不給分）

Common error：寫了兩組的相同之處（最終都全部溶血），或沒有引用數值。

(b)(i) (2)

B1 urea diffuses into the cells down its concentration gradient until its concentration inside is close to that outside (1)

B1 the cells still contain other solutes, so the water potential inside is lower than that of the solution and water enters by osmosis (1)

B1 red blood cells have no cell wall and burst after taking in water (1)

〔max. 2〕

Do not accept：尿素溶液的水勢高於細胞（2 mol dm^{-3} 溶液濃度高於細胞）

Common error：以為 *2 mol* dm^{-3} 尿素溶液是低滲溶液；須指出尿素本身進入細胞。

(b)(ii) (3)

B1 protein U allows urea to cross the cell membrane faster (a transport protein for facilitated diffusion) (1)

B1 Graph I: without protein U, haemolysis in urea solution is much slower, showing urea enters the cells much more slowly (1)

B1 Graph II: in distilled water both groups haemolyse at exactly the same rate, so the rate of water entry is unaffected by the lack of U and protein U is not a water channel (1)

〔max. 3〕

Also accept：蛋白質 U 是尿素通道／尿素轉運蛋白

Do not accept：蛋白質 U 主動轉運尿素

Common error：只根據圖 *I* 便下結論，沒有用圖 *II* 的速率排除蛋白質 *U* 運水的可能。

(b)(iii) (1)

B1 the results only support but cannot prove the conclusion, because the membranes of people lacking protein U may differ in other ways / further evidence is needed (e.g. adding protein U to membranes lacking it speeds up urea entry) (1)

〔max. 1〕

Also accept：只比較了少數志願者，樣本可能太少；使用同一批志願者的血液只可排除個體差異對兩個實驗比較的影響，不能排除其他膜蛋白的差異

Common error：只寫「要重複實驗」，沒有指出「支持」與「證明」的分別。

6. (11 marks)

(a) (3)

B1 maltase hydrolyses maltose into glucose (1)

B1 when maltase is inhibited, less glucose is produced in the small intestine / maltose cannot be absorbed (1)

B1 less glucose is absorbed into the blood capillaries of the villi, so the blood glucose level rises less (1)

〔max. 3〕

Common error：只寫「葡萄糖較少」，沒有指出吸收進入血液的步驟。

(b)(i) (2)

B1 maltase is located on the cell membrane of the epithelial cells of the small intestine, so the extract of the inner lining contains maltase (1)

B1 pancreatic juice contains pancreatic amylase but no (or very little) maltase, so it cannot digest maltose into glucose and cannot test the effect on maltase (1)

〔max. 2〕

Common error：以為胰液含所有消化碳水化合物的酶。

(b)(ii) (2)

B1 for each extract concentration, set up an extra tube with boiled enzyme extract (or water) instead of enzyme extract, keeping other conditions the same (1)

B1 subtract the glucose concentration of the corresponding control tube from that of each test tube (1)

〔max. 2〕

Also accept：在第 0 分鐘量度各試管的葡萄糖濃度，以 20 分鐘內的增加量計算結果（2 分）；另外量度各濃度提取物本身的葡萄糖濃度，再從結果中減去（2 分）

Common error：只提議「不加提取物」，這樣便無法量度抑制作用。

(c) (1)

B1 all five points plotted correctly (±0.1 mmol dm^{-3}), joined by straight lines or a smooth curve, with a key (1)

〔max. 1〕

Common error：漏了圖例或把點連到原點。

(d)(i) (1)

B1 the extract is diluted by food and digestive juices in the small intestine, so its concentration is lower (1)

B1 some of the extract is digested or absorbed in the stomach / small intestine (1)

B1 human maltase may respond to the extract differently from pig maltase (1)

〔any 1；max. 1〕

(d)(ii) (2)

B1 the extract only slows down, but does not completely stop, the digestion of maltose, so glucose is still absorbed slowly over a longer period (1)

B1 in the group without extract, glucose absorption is largely complete and insulin brings the blood glucose level back to normal, so the two groups become similar (1)

〔max. 2〕

Common error：以為提取物在 *90* 分鐘後失效，沒有提及體內平衡的調節。

7. (10 marks)

(a)(i) (1)

B1 belt transect (quadrats placed at intervals along a transect line) (1)

〔max. 1〕

Also accept：間斷樣帶 interrupted belt transect

Do not accept：只寫「樣方」；隨機取樣

(a)(ii) (2)

B1 count the number of species L in each of the five quadrats at station 4 (1)

B1 calculate the mean number per quadrat and divide by the quadrat area (0.25 m^2) to give the number per m^2 (1)

〔max. 2〕

Common error：沒有把每個樣方的數目換算成每平方米。

(b) (3)

B1 place species L in two sealed containers, one containing air only (measure oxygen with an oxygen sensor) and one filled with seawater (measure dissolved oxygen with a dissolved oxygen sensor) (1)

B1 use the same number / total mass of snails in both groups and keep the temperature constant (1)

B1 measure the decrease in oxygen concentration over a fixed time and calculate the rate of oxygen consumption per g of snail per hour for comparison (1)

B1 set up control containers without snails / repeat and take the mean (1)

〔max. 3〕

Common error：沒有方法量度氧（例如只觀察螺的活動），或沒有控制螺的質量和溫度。

(c) (1)

B1 the rate of oxygen consumption of species L in air is similar to or higher than that in seawater (1)

〔max. 1〕

Also accept：物種 L 在空氣中的氧消耗速率明顯大於零

Do not accept：物種 L 在空氣中能存活

(d) (3)

B1 support: L consumes oxygen faster in air (52) than in seawater (30), but S consumes much less in air (18) than in seawater (45) (1)

B1 support: L is mainly found at stations 4–6 near the land, which are exposed for longer at low tide, while S is concentrated at stations 1–2 (1)

B1 limitation: only oxygen consumption was measured; distribution may also be affected by other factors such as desiccation, salinity, food, predators or competition (1)

B1 limitation: laboratory results may not reflect field conditions / sample size unknown (1)

〔max. 3〕

Common error：只同意或反對，沒有同時指出證據和不足之處；評價題須兼顧兩面。

8. (8 marks)

(a) (1)

B1 most cases were adults aged 20–49 (65 cases), with the most in the 30–39 group (31 cases); very few cases were aged 1–19 (3 cases) (1)

〔max. 1〕

Do not accept：只寫「成年人較多」而沒有數據

(b) (4)

B1 the vaccine contains weakened measles virus / its antigens, which stimulate lymphocytes (B cells) to produce antibodies (primary response) (1)

B1 memory cells are formed and remain in the body for a long time (1)

B1 on later infection, memory cells trigger a secondary response, producing more antibodies more quickly to destroy the virus before symptoms appear (1)

B1 the second dose increases the proportion of people producing antibodies / boosts the response (1)

B1 when most people are immune, an infected person is unlikely to pass the virus to a susceptible person, breaking the chain of transmission (1)

B1 so infants are much less likely to come into contact with the virus (herd immunity) (1)

〔max. 4〕

Common error：只解釋 *(i)* 或只解釋 *(ii)*；兩部分各最多 *3* 分，合共上限 *4* 分。

(c) (1)

B1 only 85–93% of these age groups have protective antibodies, below 95%, so herd immunity is insufficient and the virus can spread among them (1)

〔max. 1〕

Also accept：他們兒時未有兩劑接種計劃，只接種一劑或沒有接種；他們接種疫苗多年，抗體水平隨年月下降

Common error：沒有引用圖 *II* 的百分率。

(d) (2)

B1 only 9 of 77 cases (12%) had two doses, while most of the population is vaccinated, so the proportion of vaccinated people infected is very low (1)

B1 the 1–19 age groups with the highest percentage of protective antibodies had the fewest cases (1)

B1 no vaccine is 100% effective / antibody levels of a few people may fall with time, so the claim is not valid (1)

〔max. 2〕

Common error：只表示同意或不同意，沒有引用個案數字。

9. (5 marks)

(a) (2)

B1 with the stomata kept open, fungal hyphae / spores can enter the leaf through the stomata (1)

B1 the fungus can reach the nutrient-rich mesophyll cells and absorb nutrients to grow and spread (1)

B1 the plant loses much water and is weakened, so it is less able to resist the fungus (1)

〔max. 2〕

Common error：只描述植物失水，沒有指出對真菌的好處。

(b) (3)

B1 the leaf water content of plant I fell from 86% to 55%, much more than plant H (86% to 80%) (1)

B1 the toxin keeps the stomata open, so transpiration continues and water is lost faster than it is absorbed from the dry soil (1)

B1 the cells lose water and lose turgidity, so the leaves cannot be supported and wilt (1)

〔max. 3〕

Common error：沒有引用數據，或沒有把氣孔張開連繫到失水與膨脹度。

10. (8 marks)

(a) (4)

B1 parent: blue-black, genotype Ww (since blue-black and red-brown pollen are about 1:1) (1)

B1 gametes: W and w (both pollen and ova) (1)

B1 offspring genotypes: WW : Ww : ww = 1 : 2 : 1 (1)

B1 offspring phenotypes: blue-black : red-brown = 3 : 1 (1)

〔max. 4〕

Do not accept：只畫棋盤方格而沒有文字標示親代、配子和後代

Common error：親代只寫基因型而沒有表現型，或沒有由 *1:1* 花粉比例推斷親代為雜合。

(b) (3)

B1 blue-black offspring have the genotype WW or Ww in the ratio 1 : 2 (1)

B1 only WW plants produce pollen all carrying W (all blue-black); Ww plants produce half blue-black and half red-brown pollen (1)

B1 the proportion is 1/3 (1)

〔max. 3〕

Common error：以全部後代作分母，答 *1/4*。

(c) (1)

B1 1/2 (the Ww plants) (1)

〔max. 1〕

Also accept：跟隨 (a) 的錯誤答案正確推論者可給分（error carried forward）

11. (12 marks)

Pollination, fertilisation and formation of fruits and seeds (max. 3)

B1 insects feeding on nectar carry pollen from anthers to stigmas (pollination) (1)

B1 the pollen grain grows a pollen tube and the male gamete fuses with the ovum in the ovule (fertilisation) (1)

B1 after fertilisation, ovules develop into seeds and the ovary develops into a fruit (1)

B1 with fewer pollinators, fewer ovules are fertilised, so fewer seeds and fruits form, or fruits are small or misshapen, lowering crop yield (1)

Effects on human diet and health (max. 3)

B1 fruits, vegetables and nuts become scarcer and more expensive, so people eat less of them (1)

B1 less vitamin C is taken in, which may lead to scurvy (bleeding gums, slow wound healing) (1)

B1 less dietary fibre is taken in, slowing peristalsis, causing constipation and raising the risk of colorectal cancer (1)

B1 people may eat more energy-rich processed food, giving an unbalanced diet and higher risk of obesity and cardiovascular disease (1)

B1 fewer nuts reduce a source of unsaturated fats and protein (1)

Effects on other organisms in the ecosystem (max. 3)

B1 plants and pollinators show mutualism: plants provide nectar and pollen, insects pollinate the plants (1)

B1 reproduction of wild insect-pollinated plants also falls, so the number of producers decreases (1)

B1 animals feeding on the fruits, seeds or leaves of these plants have less food and decline (1)

B1 animals that feed on pollinating insects (e.g. birds) also decline, affecting the whole food web (1)

B1 biodiversity decreases and the ecosystem becomes less stable (1)

Effective communication (3)

3: Answer is easy to follow and fluent, with no or very little irrelevant material; presented systematically with a complete structure and clear cause-and-effect. 2: Some inappropriate wording or a little irrelevant material; some repetition in organisation but structure still visible. 1: Effort is needed to understand the answer; irrelevant material obscures the main points; slightly disorganised but paragraphed. 0: Answer is incomprehensible or totally disorganised. Answers in bullet points score at most 1 mark for communication.